## SUPPLEMENTAL MATERIAL

Here we solve $3j+1=0$ to the nearest 1/4. We choose three qubits for $|j\rangle$, using two's complement with two fractional bits. So the most negative possible value is $100 \Rightarrow -1$, and the most positive possible value is $011 \Rightarrow 3/4$. We will use five auxiliary qubits in the $|a\rangle$ register, using two's complement with two fractional bits. The values that $|a\rangle$ can represent range from $10000 \Rightarrow -4$ to $01111 \Rightarrow 3\frac{3}{4}$.

We follow the same steps used for the simpler case in the text, first analyzing multiplication by 3 in Table SI. We next need to deduce an expression for each bit of $a$ in terms of the three bits of $j$. We quickly see that $a_4 = j_2$ and $a_0 = j_0$. $a_1$ is the same in the top half of the table as the bottom half, which means it does not depend on $j_2$. We see that $a_1$ is 1 whenever $j_1$ differs from $j_0$, so $a_1 = j_1 \oplus j_0$.

TABLE SI. Multiplication by 3. The first column gives the binary representation of the base-10 number in the second column. The third column multiplies by 3, and the fourth column is the binary representation of the result.

| $j$ (2's comp, $j_2.j_1j_0$) | $j$ (base 10) | $3j$ (base 10) | $3j$ (2's comp, $a_4a_3a_2.a_1a_0$) |
|---|---|---|---|
| 000 | 0 | 0 | 00000 |
| 001 | 1/4 | 3/4 | 00011 |
| 010 | 1/2 | 3/2 | 00110 |
| 011 | 3/4 | 9/4 | 01001 |
| 100 | -1 | -3 | 10100 |
| 101 | -3/4 | -9/4 | 10111 |
| 110 | -1/2 | -3/2 | 11010 |
| 111 | -1/4 | -3/4 | 11101 |

We see that $a_2$ is 1 when $j_2$ is 1 or $j_1 \cdot \overline{j_0}$ is 1, but not both, so $a_2 = j_2 \oplus j_1 \cdot \overline{j_0}$. We see that $a_3$ is 1 when $j_2 \cdot j_1 = 1$ or $j_1 \cdot j_0 = 1$, so $a_3 = j_2 \cdot j_1 + j_1 \cdot j_0$. Using Eq. (2) from the main text, $a_3 = j_2 \cdot j_1 \oplus j_1 \cdot j_0 \oplus j_2 \cdot j_1 \cdot j_0$.

Multiplication by 3 is performed by the gates before the first dashed line in Fig. S1, within the large rectangle that represents the oracle operator. The first CNOT sets $|a_4\rangle = |j_2\rangle$. The next three gates add the terms $j_2 \cdot j_1$, $j_1 \cdot j_0$, and $j_2 \cdot j_1 \cdot j_0$ in $|a_3\rangle = |j_2 \cdot j_1 \oplus j_1 \cdot j_0 \oplus j_2 \cdot j_1 \cdot j_0\rangle$.

The next gates, targeting $|a_2\rangle$, add the terms $j_2$ and $j_1 \cdot \overline{j_0}$ in $a_2 = j_2 \oplus j_1 \cdot \overline{j_0}$. The CNOT gate targeting $|a_0\rangle$ sets $|a_0\rangle = |j_0\rangle$. The final two gates set $|a_1\rangle = |j_1 \oplus j_0\rangle$.

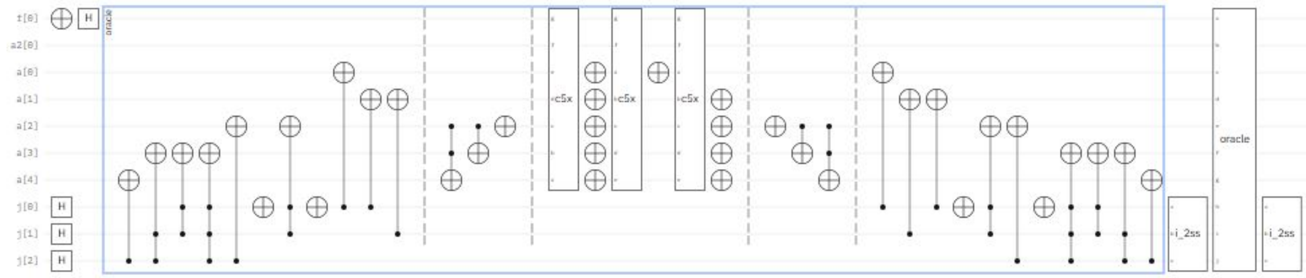


FIG. S1. A circuit to solve $3j + 1 = 0$ to the nearest 1/4.

To add 1 to $|3a\rangle$, we use the gates between the second and third dashed lines in Fig. S1. This is Fig. 1 from the main text, applied to the three integer qubits, $|a_4 a_3 a_2\rangle$. The result is correct even through $a_4$ represents $-4$, as long as the sum is within the range of five-bit two's complement with two fractional bits.

We next need to apply a NOT to the top qubit whenever $|a\rangle$ represents a value within 1/4 of 0, that is, when $a$ is 11111, 00000, or 00001. The gates required are much like those in the simpler case (solution to the nearest 1/2), but since there are five qubits in $|a\rangle$, we need five control bits. IBM Quantum does not provide a quintuply controlled NOT. We need to create a quintuply controlled NOT as shown in Fig. S2. This is used between the second and third dashed lines in Fig. S1 to flag the correct solution. Between the third and fourth dashed lines is the uncomputing step that reverses the +1 operation. The $\times 3$ operation is reversed after the fourth dashed line.

The $I - 2|S\rangle\langle S|$ operator for three qubits is a generalization of Fig. 3 in the main text. The addition of a third qubit is a little complicated because IBM Quantum does not provide a doubly controlled Z gate. We construct this gate using a Toffoli with a Hadamard on each side of the target. (This works because $Z = HXH$, as students can prove by trying it on both computational basis states.) The complete $I - 2|S\rangle\langle S|$ operator for three qubits is shown in Fig. S3.

The final complication is that we have three qubits in $|j\rangle$, so Eq. (20) from the main text indicates that we need to repeat the oracle and $I - 2|S\rangle\langle S|$ operators twice. This is shown in Fig. S1. In the IBM Quantum Composer, the simulation shows a 94.6% chance of measuring $111 \Rightarrow -1/4$, which is the solution to $3j + 1 = 0$ to the nearest 1/4.

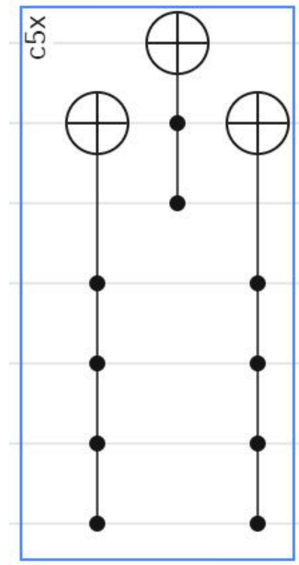


FIG. S2. Creating a quintuply controlled NOT out of two quadruply controlled NOT gates and a Toffoli gate. The top qubit is the target, and the second qubit from the top is an ancilla. The second quadruply controlled NOT returns the ancilla to $|0\rangle$ to prevent unwanted entanglement.

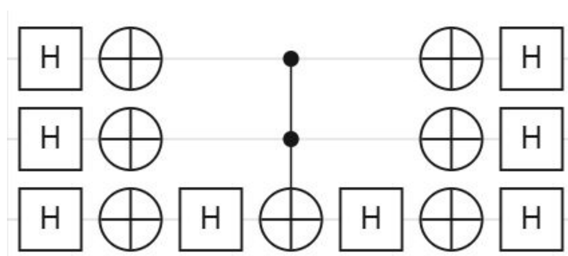


FIG. S3. The complete $I - 2|S\rangle\langle S|$ operator, using a Toffoli and two Hadamard gates as a doubly controlled Z.